

Auto-Building Spaces

Loops + functions + chat

명령 하나로 공간 전체가 저절로 지어져요

BIG PICTURE

Everything together



A loop repeats a room, a function builds it, a number decides the size.

반복 + 함수 + 채팅 숫자

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

Loop a function call

```
def on_dungeon(rooms):  
    for i in range(rooms):  
        build_room()  
        agent.move(FORWARD, 6)  
player.on_chat("dungeon", on_dungeon)
```

Type `dungeon 3` → 3 rooms, each 6 apart.

반복이 함수 호출을 여러 번 실행해요

CONCEPT

Functions inside functions

```
def build_room():  
    build_wall(5)  
    build_wall(5)  
    build_wall(5)  
    build_wall(5)
```

dungeon 2 → 2 rooms × 4 walls = **8 walls**.

방 함수가 벽 함수를 재사용해요

CONCEPT

Randomness makes it feel alive

```
def on_towers(count):  
    for i in range(count):  
        build_tower(randint(3, 8))  
        agent.move(FORWARD, 4)  
player.on_chat("towers", on_towers)
```

Each tower gets a random height — no two the same.

랜덤 높이로 매번 다른 풍경

WATCH IT!

Loop builds room by room

```
room = 1
```

```
room <= rooms
```

```
build_room(); move(FORWARD, 6)
```

```
▸ Built
```

반복이 끝날 때까지 방을 지어요

PREDICT 🤔

How many walls?

Predict first — then click reveal

```
def build_room():  
    build_wall(5)  
    build_wall(5)  
    build_wall(5)  
    build_wall(5)  
  
player.on_chat("dungeon", on_dungeon)  
# you type:  dungeon 2
```

Reveal output 🔍 reveal

VOCAB

New words today

argument the number you send with a command

range how many times a loop repeats

randint a random number in a range

offset how far to move between pieces

인수

범위

난수

간격



What's wrong with this code?

Goal: `dungeon 5` builds 5 rooms — but always builds just **one**.

buggy

```
def on_dungeon(rooms):  
    for i in range(1):  
        build_room()  
        agent.move(FORWARD, 6)  
player.on_chat("dungeon", on_dungeon)
```

The argument `rooms` is never used.



DEBUG

The bug

Bug: the loop is `range(1)`, ignoring `rooms`. Loop over the argument.

fixed

```
def on_dungeon(rooms):  
    for i in range(rooms):  
        build_room()  
        agent.move(FORWARD, 6)  
player.on_chat("dungeon", on_dungeon)
```

YOUR TURN!

Auto-build a space

✓ Chat command with a number argument

✓ A loop repeats a room piece

✓ Use the argument in `range`

✓ Move the right distance between pieces

✓ Try a small number, then a big one



QUIZ

QUESTION 1 OF 3

`dungeon 5` only builds one room. Why?

5 is too many rooms.

`build_room` is misspelled.

The loop is `range(1)` and ignores `rooms`.

Chat commands only build once.



QUIZ

QUESTION 2 OF 3

Rooms are 6 long but the agent moves 5 between them. What breaks?

The loop never ends.

Nothing — 5 is correct.

The rooms get too tall.

Each room overlaps the back wall of the last — move 6.



QUIZ

QUESTION 3 OF 3

Why do `build_tower(randint(3, 8))` towers all differ?

`randint` picks a new random height each loop.

Towers always shrink.

The agent gets tired.

`range` adds 1 each time.

QUIZ

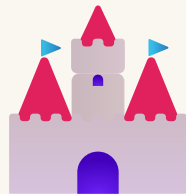
Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



A whole space, one command!

Next: build your final game.

다음 주: 나만의 미니게임 완성